

PharmTwinAI

AI-Powered Pharmacy Digital Twin — Colour Student Teaching Pack
Modules 0–11 · Architecture · Formulae · Worked Numericals · Viva Bank

Industry context: Bhagyashree Medical (Juinagar, Navi Mumbai) — retail pharmacy operations

DEV data mode: explicitly labeled **dev_synthetic** — NOT live sponsor POS sales

Stack: Python tkinter desktop + MySQL 8 · Offline pipeline Steps 1–5 · services/ online layer

MODULE 0 — COVER: COLOUR LEGEND, HOW TO STUDY, VIVA MARKING

0.1 Colour pedagogy (same as MU PLM colour notes)

Colour	What it marks	How you should use it
Navy	Module banners, section spine, table headers	Scan first — structural map of the pack
Crimson	Definitions and key terms	Quote in viva / report; examiners look here first
Green	Formulae and exact identities	Reproduce exactly — WMAPE, MASE, SS, ROP, cosine
Purple	Worked numericals / mini examples	Practise end-to-end; easy viva points
Orange	Comparison tables & exam cautions	"Differentiate..." traps and common mistakes
Bordered panels	ASCII flowcharts / architecture traces	Redraw in ~90 seconds in a demo or viva

0.2 How to study this pack (fresher plan)

Treat this like a laboratory teaching pack, not a novel. First pass: read every navy banner and every crimson definition box — that alone covers the viva vocabulary. Second pass: copy every green formula onto one A4 sheet (Module 9 has a summary page). Third pass: solve every purple numerical without looking, then check. Fourth pass: redraw the bordered diagrams from memory. Fifth pass: answer Module 10 viva hints out loud in 30–45 seconds each.

Day	Focus	Deliverable you should be able to produce
1	Modules 0–2	Twin definition + D1–D8 table + architecture redraw
2	Modules 3–4	Catalog vs working stock + Croston/SBA/TSB + WMAPE/MASE
3	Module 5	Three SS/ROP numericals + FEFO vs FIFO + sim policies
4	Modules 6–7	Gate logic + every desktop page walkthrough
5	Modules 8–11	Pipeline I/O + glossary + 40 viva Qs + limitations honesty

0.3 Viva / evaluation marking scheme (what separates marks)

Four points examiners look for — where marks separate

- Named methods** — Croston, SBA (Syntetos–Boylan), TSB (Teunter–Syntetos–Babai), LightGBM pinball quantiles, FEFO, TF-IDF cosine — say the full names once.
- Green formulae written correctly** — WMAPE, MASE, SS, ROP, σ from quantiles, cosine. Wrong MAPE claim = automatic trap.
- One concrete pharmacy example** — Navi Mumbai monsoon rain > 20 mm, PM2.5 winter, Schedule H1 register duty, FEFO near-expiry lot.
- Honesty boundaries** — data is **dev_synthetic**; anemia screening is gated (D7); substitutes are pharmacist-reviewed not auto-prescribe (D4); env features are research (D5).

Band	Typical evidence	What goes wrong
High	Formula + numerical + diagram + limitation sentence	Overclaims rare
Mid	Correct vocabulary but no numerical / no diagram	Vague "AI predicts demand"
Low	Calls twin a chatbot / claims real POS data / uses MAPE	Examiner trap hit

0.4 What NOT to claim (honesty sheet)

CAUTION / EXAMINER TRAP — Never claim synthetic DEV transactions are real Bhagyashree Medical POS sales. Never claim the system auto-prescribes or diagnoses. Never claim weather/PM2.5 causally proven on sponsor data (D5). Never mutate the ~253,973 reference catalog from the Inventory UI. Never use MAPE as the headline metric for intermittent demand.

Definition — What PharmTwinAI is (one sentence for cover / abstract)

A desktop pharmacy digital twin that mirrors retail inventory and intermittent demand state in MySQL, forecasts with Croston-family and LightGBM quantiles (WMAPE/MASE), computes probabilistic safety stock and FEFO policies in isolated simulation, and suggests legally gated medicine substitutes for pharmacist review — using explicitly labeled synthetic DEV data until sponsor POS arrives.

0.5 Pack map (Modules 0–11)

Module	Title	Must-know artefact
0	Cover / study / marking	Colour legend + honesty sheet
1	Problem & twin definition	D1–D8 + twin vs dashboard table
2	Architecture	UI→services→DB path + folder map
3	Data foundation	253,973 vs ~3,000 + clone flow
4	Forecasting	Croston/SBA/TSB steps + WMAPE/MASE
5	Inventory science	SS/ROP numericals + FEFO + sim
6	Substitutes	TF-IDF + H1/AWaRe/CDSCO gates
7	Desktop & ops	Every page + failure modes
8	Pipeline Steps 1–5	Diagram + outputs per step
9	Glossary & formula sheet	Large glossary + green summary
10	Viva bank (40+)	Short model-answer hints
11	Limitations & ethics	D7 anemia gate + future work

1.1 Why a retail pharmacy needs this system

A neighbourhood medical store (example context: Bhagyashree Medical, Juinagar, Navi Mumbai) faces four recurring operational pains: (a) intermittent SKU demand — many weeks with zero sales then a spike; (b) dated lots that expire and become waste if picked poorly; (c) stockouts that lose sales and push patients elsewhere; (d) pressure to suggest an alternative when a brand is out, without breaking Schedule H1 / antibiotic stewardship / banned FDC rules. Classic “dashboard-only” software shows yesterday’s numbers but does not hold a what-if twin, probabilistic buffers, or gated substitutes.

Definition — Pharmacy Digital Twin (PharmTwinAI)

A living software mirror of a retail pharmacy's inventory, demand and decision state — kept in sync with operational data (MySQL), able to forecast intermittent sales, recommend reorders with explainable safety stock, run what-if simulations without touching live stock, and suggest legally gated medicine substitutes for pharmacist review.

Four points examiners look for — unpacking the twin definition

1. **Business system, not a chatbot.** Inventory truth + forecasts + policy simulation.
2. **Entire stock lifecycle of lots** — receipt, sale, expiry write-off, reorder — not just a drug list.
3. **Manages information and decisions** about stock, not clinical diagnosis.
4. **Regulatory gates on substitutes** — Schedule H1 / WHO AWaRe / CDSCO — human pharmacist remains in the loop.

1.2 Stakeholders and what each needs

Stakeholder	Need	How PharmTwinAI addresses it
Retail pharmacist	Know what to reorder & what expires	Recs + alerts + FEFO lots
Store owner	Reduce waste & stockouts	Policy sim fill-rate / waste KPIs
Project team / college	Reproducible offline science	Steps 1–5 scripts + metrics
Industry sponsor	Safe demo without live POS leak	dev_synthetic label + D3
Examiner / viva panel	Named methods + honesty	This pack + acceptance checks
Patient (indirect)	Availability + legal dispensing	Stockouts→substitutes review UI

1.3 Scope — in vs out

In scope (built / demoable)	Out of scope / gated
Desktop twin UI, MySQL stock, Steps 1–5 offline pipeline, intermittent FC, SS/ROP, FEFO sims, gated substitutes, inventory clone CRUD	Live sponsor POS ingest (until export), clinical diagnosis, auto-prescribe, paid cloud deploy (D6), anemia screening product (D7), other screening (D8)

1.4 Locked design decisions D1–D8 (memorise with reasons)

These are project locks documented in **docs/decisions_locked.md**. In a viva, naming the ID (D2, D3...) plus a one-line reason scores higher than vague “we chose desktop”.

ID	Decision	Explanation / viva reason
D1	Evolve in place under Agent/; restructure into appl/, db/, services/	Keep one repo; clear layers for UI / schema / business logic
D2	Native desktop app (tkinter now; PySide6 optional later) — not Streamlit-first	Pharmacy software culture expects an installable desktop tool
D3	No sponsor export yet → build explicitly labeled synthetic data	Academic honesty + NDA; banner shows SYNTHETIC DEV DATA
D4	Pharmacist-reviewed inventory alternatives only (gated)	Not clinical auto-prescribe; human remains accountable
D5	Weather / epi covariates = research/DEV until proven on real sales	Do not overclaim causality from monsoon/PM2.5 features
D6	Local / college lab only (no paid cloud for ~6 months)	Cost and data-control constraint

ID	Decision	Explanation / viva reason
D7	Anemia: research docs only; no product implementation until core acceptance	Scope control — core twin must land first
D8	Other screening: research-only until per-condition approval	Avoid feature creep into clinical product claims

CAUTION / EXAMINER TRAP — If asked “is this an anemia detection project?” — answer No for the shipped product (D7). Core acceptance is inventory twin + forecast + FEFO + gated substitutes.

1.5 Goals mapped to features

Goal	In plain words	Feature in PharmTwinAI
Stock truth	What is on the shelf; which lot dies first?	Inventory + FEFO lots
Demand foresight	How many units next week under uncertainty?	Croston/SBA/TSB + LightGBM q50/q90/q95
Reorder science	How much buffer against lead-time risk?	$SS = Z \cdot \sigma \cdot \sqrt{L+R}$, ROP
Safe experimentation	What if demand $\times 1.2$ in monsoon?	Isolated simulations
Stockout recovery	What else can we dispense legally?	TF-IDF substitutes + gates

1.6 Comparison — Digital twin vs dashboard vs plain inventory software

Basis	Plain inventory SW	Analytics dashboard	PharmTwinAI digital twin
State mirror	Current stock list	Charts of past KPIs	Stock + demand + decision state
Forecasting	Rare / manual	Maybe external BI	Built-in intermittent + quantiles
What-if	Usually none	Filters only	Cloned sim; live stock untouched
Safety stock science	Rule-of-thumb	Hidden in spreadsheet	Explainable SS/ROP in recs
Substitutes	Staff memory	None	TF-IDF + H1/AWaRe/CDSCO gates
Data honesty	Live POS assumed	Live POS assumed	dev_synthetic labeled (D3)

Definition — Twin isolation invariant

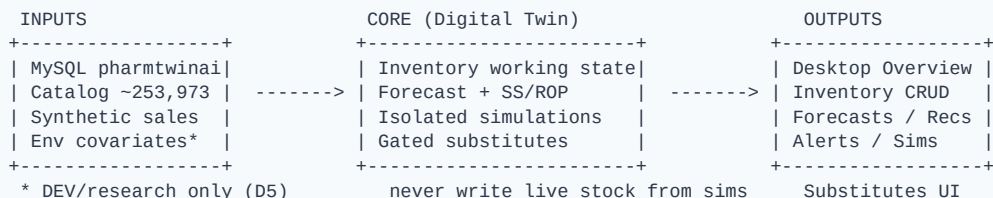
A simulation or what-if run must never mutate live medicine_batches quantities. The project fingerprints SUM(qty_on_hand) before/after; if it changes, isolation is broken.

MODULE 2 — SYSTEM ARCHITECTURE

2.1 Full stack at a glance

PharmTwinAI splits into an **offline science pipeline** (Steps 1–5 under src/ + scripts/) that writes CSV/Parquet under data/processed/, and an **online desktop twin** (app/main.py + services/) that reads/writes MySQL database **pharmtwinai**. Heavy LightGBM training stays offline; the UI never trains models live.

Architecture trace — Inputs → Twin → Desktop outputs



2.2 Technology stack with fresher rationale

Layer	Choice	Fresher rationale
DB	MySQL 8 localhost	ACID stock + sales; familiar college stack
UI	Python tkinter desktop (D2)	Installable; pharmacy-software expectation
Services	services/*.py in-process	UI calls Python helpers; no separate API server required
Science	src/ offline Steps 1–5	Reproducible ML; seed=42 default
Models	LightGBM quantile regressors	q50 point + q90/q95 for σ
NLP	sklearn TF-IDF + cosine	Composition text similarity for substitutes
Config	.env + app_meta	DB password; data_mode=dev_synthetic

2.3 Offline vs online split

Concern	Offline (scripts/src)	Online (app/services/MySQL)
Train LightGBM	Yes (Step 3)	No — reads forecasts table
Generate synthetic Tx	Yes (Step 1)	No
Edit working stock	Seed only	Yes — inventory_write guards
Run FEFO policy grid	Yes (Step 4)	Import cached / what-if clone
Substitute gates	Step 5 audit CSVs	Live lookup in Substitutes page

2.4 Request path — UI → services → DB

Request path redraw

```
[ttk Page in app/main.py]
  | user clicks Refresh / Add lot / Lookup substitute
  v
[services/*.py] -- SQL / business rules --> [MySQL pharmtwinai]
  | e.g. inventory_write.add_medicine() | medicines, medicine_batches,
  | forecasts.list_forecasts()         | forecasts, recommendations,
  | substitutes.suggest()               | alerts, simulation_runs, ...
  v                                   v
[UI table / messagebox]             [inventory_audit on mutations]
```

There is no separate REST microservice in the default demo. The desktop process imports services modules directly. That simplifies college-lab install (D6) but means the PC running the UI must reach localhost MySQL.

2.5 MySQL main tables — what each holds

Table	Purpose	Exam one-liner
medicines	Catalog + working SKUs	source_system separates reference vs shelf
medicine_batches	Lots with qty/mfg/expiry	FEFO picks earliest expiry

Table	Purpose	Exam one-liner
sales_transactions / sales_items	Tx header + lines	unmet_qty captures stockout demand
forecasts	yhat / bounds / model	Loaded by load_analytics from Step 3
recommendations	REORDER etc + SS fields	explanation_text shows formula
simulation_runs / results	What-if metadata + KPIs	Must not touch batches
alerts	LOW_STOCK / EXPIRY...	Near-expiry $\leq 90d$ typical
inventory_audit	ADD/REMOVE trail	Migration 002
app_meta	data_mode flag	dev_synthetic banner
digital_twin_snapshots	Sync hash / state_json	Staleness signal

2.6 Folder map — purpose of major paths

Path	Role
src/	Catalog, covariates, Croston/SBA/TSB, LightGBM, FEFO, substitutes, regulatory
services/	Twin summary, inventory R/W, forecasts, alerts, simulations, substitutes
app/main.py	Desktop navigation pages (Overview...Settings)
db/migrations/	001 schema; 002 form_type/qty_unit/pharmacy/audit
scripts/	run_step1..5, seed_mysql, load_analytics, snapshot_originals, migrate, demo
data/originals/	Frozen pipeline copies — do not edit
data/processed/	Working pipeline outputs (overwriteable)
docs/	architecture, decisions_locked, acceptance_checklist, data dictionary

2.6.1 Important src/ files (name in viva)

File	Purpose
src/catalog.py	Dedupe India A–Z catalog; stratified assortment sampling
src/generate_transactions.py	Step 1 synthetic intermittent sales + batches
src/forecast_intermittent.py	Croston, SBA, TSB, moving average
src/metrics.py	WMAPE, MASE (MAPE excluded on purpose)
src/forecast_env_lgbm.py	LightGBM quantile q50/q90/q95
src/safety_stock.py	σ from quantiles; SS; ROP; order-up-to
src/inventory_fefo.py	FEFO/FIFO allocate; OTC markdown tiers
src/substitutes_nlp.py	TF-IDF engine + gate pipeline
src/regulatory/*.py	Schedule H1, WHO AWaRe, CDSCO banned FDC

2.6.2 Important services/ files

File	Purpose
services/db.py	MySQL connect from .env; probe
services/twin.py	Catalog vs stocked counts; data_mode banner
services/inventory.py	List stocked medicines / lots
services/inventory_write.py	Guarded add/remove; block reference mutation
services/forecasts.py	Read forecasts table
services/recommendations.py	Reorder recommendations
services/alerts.py	Expiry / low-stock alerts
services/simulations.py	Cached Step 4 import; what-if; isolation
services/substitutes.py	Live gated substitute lookup

CAUTION / EXAMINER TRAP — Do not tell the examiner that Streamlit is the primary UI. D2 locks desktop-first. Streamlit may appear in older notes as a rejected option.

2B.1 Why MySQL and not only Parquet files for the twin?

Parquet/CSV under data/processed/ are excellent for reproducible offline science: you can re-run Step 3 and diff metrics. But a retail twin needs concurrent-safe stock mutations, foreign keys from batches→medicines, audit rows, and a queryable forecasts table for the desktop. MySQL gives ACID updates when a pharmacist adds a lot or when seed loads sales_items. The twin's "live state" is therefore MySQL; the pipeline's "scientific artefacts" remain files. load_analytics bridges them.

Definition — Dual store pattern

Offline artefacts (Parquet/CSV/JSON summaries) + online operational store (MySQL). Desktop reads MySQL; Steps 1–5 write files; seed/load_analytics copy science into MySQL.

2B.2 sales_items.unmet_qty — why examiners care

If you only record fulfilled qty, a stockout looks like "zero demand", so the forecast learns to predict zero forever — a classic retail analytics bug. unmet_qty stores the portion of demand that could not be sold. Step 2/3 demand = qty + unmet_qty. In viva, draw a tiny table: demand 10, stock 4 → qty=4, unmet=6.

WORKED MINI — unmet_qty

Customer wants 10 strips; on-hand 4 across lots.

sales_items: qty=4, unmet_qty=6. Revenue on 4 only.

Weekly demand used in FC: 10, not 4. Safety stock will not collapse falsely.

2B.3 app_meta and the synthetic banner

001_init_schema seeds app_meta with data_mode='dev_synthetic'. services/twin.py reads this for the Overview banner: SYNTHETIC DEV DATA — not Bhagyashree Medical sales. If a future sponsor load arrives, the mode string changes — the UI honesty signal must follow the data, not a hardcoded slide.

2B.4 inventory_write guards (name the error)

Attempt	Guard behaviour
UPDATE/DELETE medicines where source_system='reference'	Blocked — InventoryGuardError
Add lot on reference medicine_id	Blocked — must clone first
Add medicine from catalog search	INSERT new working row + optional lot
Remove working lot	Allowed + inventory_audit row

2B.5 Simulation service contract

Simulation service contract

```
Desktop Simulations page
-> services/simulations.py
    A) import cached Step 4 summary into simulation_runs/results
    B) what-if: clone in-memory / scenario_json demand shock
-> NEVER UPDATE medicine_batches.qty_on_hand
-> optional fingerprint: SELECT SUM(qty_on_hand) before/after
```

2B.6 Digital twin snapshots

digital_twin_snapshots stores synced_at, source_hash, is_stale, state_json. Purpose: tell the Overview whether analytics are fresh relative to inventory mutations. You do not need to memorise every JSON key — memorise the idea: staleness is a first-class twin concept, not a vague "refresh the page".

Four points examiners look for — architecture viva pack

1. Say dual store: files for science, MySQL for twin state.
2. Say unmet_qty prevents false zero demand.
3. Name InventoryGuardError for reference immutability.
4. Say simulations must fingerprint live stock.

3.1 Three layers of “medicine data”

Definition — Reference catalog vs working inventory vs DEV synthetic sales

The India medicines knowledge base (~253,973 deduped SKUs, `source_system='reference'`) is immutable. The pharmacy's sellable shelf is a separate working set (`source_system ∈ {dev_synthetic, pharmacy, sponsor}`) with lots carrying quantity, manufacturing date and expiry. Sales used in DEV are generated synthetically and labeled — they are not Bhagyashree POS exports.

Store	Approx. size	Mutable?	Purpose
reference medicines	~253,973	NO	Search / clone / NLP knowledge
working medicines	~3,000	YES	Shelf assortment
medicine_batches	thousands	YES*	Qty + mfg + expiry (*working meds only)
sales_* (DEV)	~254k Tx header scale	seeded	Intermittent demand + unmet_qty
data/originals/	file snapshot	NO	Integrity freeze of pipeline inputs

3.2 Kaggle / India A–Z → dedupe → assortment

External catalog path (not committed in the git repo): an Extensive A–Z medicines dataset of India (~2.5L+ rows). Step 1 loads and dedupes to **253,973** unique reference SKUs (`catalog_deduped.parquet`). Then stratified sampling by demand cohort / therapeutic tags builds an active retail assortment of **~3,000** SKUs (`assortment_3k.csv`). Cohorts matter because intermittent slow-movers dominate pharmacy shelves; sampling must not keep only fast movers.

Catalog → assortment → synthetic Tx

```
[India A-Z CSV ~2.5L+]
  | load + normalize names/composition keys
  v
[catalog_deduped.parquet] ----- 253,973 reference SKUs
  | stratified sample by cohort / class
  v
[assortment_3k.csv] ----- ~3,000 working SKUs
  | + covariates_navi_mumbai + Bernoulli/Poisson sales process
  v
[transactions_synthetic + batches_fefo] --> seed_mysql (reference + working)
```

3.3 Synthetic transaction generation idea

Retail pharmacy demand is intermittent: most SKU–days are zero. The generator (Step 1) builds a daily panel conditioned on covariates (day-of-week, monsoon rain flags, PM2.5 spikes, health-alert flags) and SKU cohort. Conceptually: a Bernoulli (or sparse point-process) draw decides whether a demand event occurs; conditional on an event, a Poisson (or similar count) draw decides units. When on-hand is insufficient, unmet demand is recorded (`unmet_qty`) so forecasting trains on latent demand, not only fulfilled sales.

Four points examiners look for — synthetic data honesty

1. Always say **dev_synthetic** — never “our pharmacy’s real sales”.
2. Purpose: unblock ML + UI until sponsor POS export arrives (D3).
3. Seed default 42 → reproducible demos / acceptance.
4. Covariates are Navi Mumbai–style synthetic weather/epi series for research features (D5).

3.4 Batches and FEFO fields

Field	Meaning	Why FEFO needs it
batch_no	Lot identifier	Traceability
mfg_date	Manufacturing date	Secondary sort / audits
expiry_date	Use-before date	Primary FEFO sort key
qty_on_hand	Sellable units in lot	Allocation depletes this
unit_cost	Cost basis	COGS / waste valuation in sim
qty_unit	TABLETS/ML/...	Display + form consistency (002)

Field	Meaning	Why FEFO needs it
received_at	Receipt timestamp	FIFO sort key

3.5 form_type and qty_unit

Form (type)	Default qty unit	Quantity means
TABLET / CAPSULE	TABLETS / CAPSULES	Count of units
SYRUP / DROPS	ML	Volume
INJECTION	VIALS	Number of vials/amps
CREAM / INHALER	UNITS / PACKS	Pack count

Migration 002 adds form_type, qty_unit, cloned_from_medicine_id on medicines, qty_unit on batches, pharmacy in source_system enum, and inventory_audit. Apply with: py -3 scripts/apply_migration_002.py

3.6 Clone flow — add from catalog without mutating reference

Working-copy add flow

```

Add-from-catalog (safe clone):
  [Search reference] --copy fields--> [INSERT pharmacy/dev_synthetic medicine]
      |                               |
      |                               +--> [INSERT lot: qty, mfg, expiry]
      |
      +---- NEVER UPDATE/DELETE reference row ----+
              InventoryGuardError if attempted

```

CAUTION / EXAMINER TRAP — Inventory "Add medicine" CLONES from catalog into a new working row. Deleting or editing a reference row is blocked. Favourite viva trap.

3.7 Originals snapshot

scripts/snapshot_originals.py copies critical processed artefacts into data/originals/ as a frozen integrity baseline. Students / demos should not hand-edit originals. Use --force only when intentionally refreshing the freeze after a deliberate re-pipeline.

3.8 Schema fields students should recognise

3.8.1 medicines (selected)

Field	Notes
medicine_id	PK
name / generic_name	Display + search
rx_schedule	OTC / H / H1 / X / UNKNOWN
demand_cohort	Sampling / metrics slices
source_system	reference dev_synthetic pharmacy sponsor
form_type / qty_unit	Added in 002
cloned_from_medicine_id	Lineage to reference
unit_mrp	Pricing display / markdown base

3.8.2 forecasts / recommendations / simulation_runs

Table.field	Meaning
forecasts.yhat / yhat_lower / yhat_upper	Point and interval-ish bounds
forecasts.model_name	e.g. lgbm_env / sba
forecasts.metrics_json	WMAPE/MASE crumbs
recommendations.action_type	REORDER / HOLD / REVIEW_EXPIRY / ...
recommendations.safety_stock / reorder_point	From SS maths
recommendations.explanation_text	Human-readable formula narrative
simulation_runs.scenario_json	What-if knobs
simulation_results.metrics_json	Fill rate, waste, etc.

MODULE 3B — DATA DEEP DIVE

3B.1 Cohort-aware assortment sampling

src/catalog.py does not take the first 3,000 CSV rows. It stratifies so that slow-moving and fast-moving cohorts both appear. If you sampled only top sellers, Croston would look unnecessary and FEFO waste would be understated. Demand_cohort tags later support metric slicing (intermittent vs smooth).

3B.2 Covariate series construction (Navi Mumbai style)

Series	Construction idea	Use
Daily rainfall	Synthetic monsoon peaks; flag >20 mm	Lags in LightGBM
PM2.5	Winter elevation pattern	Respiratory-related features
Health alerts	Sparse binary flags	Short demand bursts
Calendar	DOW / week index	Routine structure

CAUTION / EXAMINER TRAP — These are engineered DEV series for method plumbing (D5), not scraped BMC claims. Do not cite fake municipal datasets.

3B.3 Batch generation aligned to FEFO

Step 1 writes batches_fefo with staggered expiry dates so the simulator has something interesting to pick. If every lot shared one expiry, FEFO=FIFO and the policy comparison would be vacuous. Multiple lots per SKU are intentional teaching structure.

3B.4 Clone lineage field

Definition — cloned_from_medicine_id

When Inventory clones a reference medicine into a working row, cloned_from_medicine_id stores the lineage. Useful for audits (“where did this shelf SKU come from?”) without mutating the reference row.

3B.5 stock_movements ledger idea

The schema includes stock_movements with movement_type $\in$ {SALE, PURCHASE, RETURN_IN, RETURN_OUT, ADJUST, WRITEOFF_EXPIRY, TRANSFER}. Even if the demo path emphasises batches qty_on_hand, the ledger concept is how a production twin would explain every unit change. Mention it if asked “how do you audit stock?”.

3B.6 demand_coverage statuses

Status	Meaning
OBSERVED	Normal observed demand day
STORE_CLOSED	No trading — do not treat as zero demand naively
MISSING_EXPORT	Gap in feed (future sponsor case)
STOCKOUT_UNMET	Demand existed but could not be filled

WORKED MINI — why STORE_CLOSED $\neq$ zero demand

If Sunday closed is stored as y=0 without a coverage flag, models learn false zeros.

Coverage status lets science distinguish “no customers” vs “shop shut” vs “stockout”.

DEV synthetic path still teaches the schema even when calendar is simplified.

MODULE 4 — INTERMITTENT DEMAND FORECASTING

4.1 Why pharmacy demand is intermittent

Definition — Intermittent demand

A demand series with many zeros (no sale that week) and occasional positive spikes. Typical of slow-moving pharmacy SKUs. Classic MAPE fails because division by zero (or near-zero) actuals is undefined / explosive.

In a 3,000-SKU neighbourhood assortment, antibiotics and chronic meds may move weekly, but many dermatology / specialty packs sit idle for weeks. Forecasting must (1) model the probability of a non-zero week and (2) model the size when demand occurs. That is exactly why Croston-style methods exist.

4.2 ADI / CV² demand patterns (Syntetos–Boylan)

Average Demand Interval (ADI) measures how many periods between non-zero demands. CV² is the squared coefficient of variation of demand sizes. Crossing literature thresholds (commonly ADI=1.32 and CV²=0.49 in the classic SB classification) yields four patterns. PharmTwinAI reports metrics by pattern slice so a method that “wins overall” is not hiding failure on lumpy SKUs.

Pattern	ADI & CV ² idea	Why slice metrics	Method intuition
Smooth	Frequent demand, steady size	Easier baselines	MA can compete
Erratic	Frequent, wild sizes	Variance-driven error	Quantiles help tails
Intermittent	Long gaps, steady size	Croston family shines	SBA/TSB strong
Lumpy	Long gaps + wild sizes	Hardest; report separately	Expect higher WMAPE

4.3 Croston — step-by-step

Croston (1972 idea family): do not smooth the raw zero-inflated series as if it were continuous. Instead maintain two smoothed states updated only when demand occurs.

Croston update sketch

```
On non-zero demand at time t (size y_t, periods since last demand = q):
  z_t = z_{t-1} + α (y_t - z_{t-1})    # smoothed demand size
  p_t = p_{t-1} + α (q - p_{t-1})      # smoothed inter-demand interval
Forecast for next period: yhat = z_t / p_t
On zero demand: do not update z,p; still emit yhat = z/p
```

Croston point forecast: $\blacksquare = z / p$

z = smoothed demand size; p = smoothed interval between demands; α typically small (e.g. 0.1).

4.4 SBA — Syntetos–Boylan Approximation

Croston's $\blacksquare = z/p$ is known to be positively biased for intermittent series. Syntetos–Boylan Approximation applies a correction factor using the same α :

SBA: $\blacksquare = (1 - \alpha/2) \times (z / p)$

Same z,p updates as Croston; multiply by $(1 - \alpha/2)$. Implemented in `src/forecast_intermittent.py`.

4.5 TSB — Teunter–Syntetos–Babai

TSB updates a demand **probability** π every period (including zeros), and a size z on positive demands. Forecast = $\pi \times z$. This reacts faster after long zero runs because π decays toward zero when no demand arrives.

TSB update sketch

```
Each period:
  if y_t > 0: z <- z + α_size (y_t - z)
              π <- π + α_prob (1 - π)
  else:      π <- π + α_prob (0 - π)    # probability decays
              z unchanged
   $\blacksquare = \pi * z$ 
```

4.6 Moving average baseline

Trailing mean over a fixed window (project uses an 8-week style MA). Simple, transparent, often surprisingly competitive on smooth SKUs, weak on long zero gaps (it smears zeros and spikes together).

4.7 Mandatory metrics — WMAPE and MASE

WMAPE = $\sum |y - \blacksquare| / \sum |y|$

MASE = $\text{mean}(|y - \blacksquare|) / \text{mean}(|y_t - y_{t-1}|)$ [naive seasonal/lag-1 scale]

Project rule: evaluate with WMAPE and MASE only — never MAPE on intermittent retail series. Demand y includes unmet_qty so stockouts do not hide true demand.

WMAPE weights errors by total absolute demand — a miss on a high-volume week hurts more than on a tiny week, and zeros in the denominator of classical MAPE are avoided because the denominator is the sum across the evaluation window. MASE scales error by a naive benchmark (lag-1 absolute changes), so a score < 1 means “better than naive”.

4.8 Why MAPE fails — purple numerical

WORKED NUMERICAL — MAPE blow-up on intermittent demand

Week demands $y = [0, 0, 0, 10]$. Forecasts $\hat{y} = [1, 1, 1, 8]$.

Classical MAPE averages $|y - \hat{y}|/|y|$ only on weeks with $y \neq 0$, or worse, tries to divide by $y=0$.

If someone naively uses $\epsilon=0.01$ instead of 0 for zeros: errors $\approx |0-1|/0.01 = 100$ (×3 weeks) + $|10-8|/10 = 0.2 \rightarrow$ absurd “MAPE”.

WMAPE = $(|0-1|+|0-1|+|0-1|+|10-8|) / (0+0+0+10) = (1+1+1+2)/10 = 0.50$ (50%).

Moral: report WMAPE/MASE; never headline MAPE for this project.

CAUTION / EXAMINER TRAP — Examiner trap: “Why not MAPE?” — Answer: zeros make MAPE undefined/unstable; project metrics module intentionally excludes MAPE.

4.9 Train / test weekly panel

Step 2/3 build a dense weekly panel (missing weeks filled with zero demand). Default split ~75% train / 25% test by time (not random SKU shuffle — that would leak future). Rolling one-step-ahead evaluation is used for intermittent baselines. Demand definition: qty + unmet_qty.

Weekly panel flow

```
Daily sales_items (qty, unmet_qty)
  | aggregate to week per medicine_id
  v
Dense weekly panel (zeros filled)
  | time split ~75/25
  v
Train intermittent / LGBM --> score WMAPE+MASE on test weeks
  |
  +--> step2*.csv / step3_forecasts_weekly.parquet
```

4.10 LightGBM quantile / pinball loss

Definition — Pinball (quantile) loss

A loss that asymmetrically penalises over- vs under-prediction so the model converges to a conditional quantile (e.g. 0.5, 0.9, 0.95) rather than only the mean.

PharmTwinAI trains three LightGBM regressors with objective='quantile' at $\alpha \in \{0.5, 0.9, 0.95\}$. q50 is the planning point forecast; q90/q95 feed σ estimation for safety stock. Ablation: lgbm_env vs lgbm_noenv compares environmental features.

4.11 Environmental features (D5 caution)

Feature family	Example	Intuition
Rainfall	rain > 20 mm day flag; 7/14-day lags	Monsoon footfall / illness patterns
Air quality	PM2.5 spikes (winter)	Respiratory OTC / related demand
Health alerts	civic alert flags	Short bursts of related SKUs
Calendar	DOW / week-of-year	Routine refill rhythms
Demand lags	lag-1..k weekly qty	Autoregression backbone

CAUTION / EXAMINER TRAP — D5: env covariates are research/DEV until validated on real sponsor sales. You may say “we engineered monsoon and PM2.5 features”; do NOT say “we proved rain causes sales”.

4.12 Feature importance interpretation

Step 3 writes step3_feature_importance.csv. High importance on demand lags is expected. Non-trivial importance on rain/PM2.5 is interesting but not causal proof. In viva: “importance $\neq$ causation; D5 applies.”

4.13 How results land in MySQL forecasts

After Steps 2–3, `scripts/load_analytics.py` (or full `seed_mysql`) inserts rows into forecasts with grain `W`, `yhat` from `q50` (or chosen model), bounds from quantiles, `model_name`, `trained_through`, `metrics_json`. Desktop Forecasts page reads this table — it does not retrain.

WORKED MINI-EXAMPLE — Reading a quantile forecast

Suppose for SKU A next week: $q50 = 8$, $q90 = 18$, $q95 = 24$ (units).

Point forecast for planning ≈ 8 . Upper stress ≈ 24 .

Rough σ from $(q95 - q50) / Z_{0.95} \approx (24 - 8) / 1.645 \approx 9.7$ units (then blended with `q90` spread in code — see Module 5).

This σ feeds the safety-stock formula.

4B.1 Croston worked numerical (hand calculation)

WORKED NUMERICAL — Croston one update

Start: $z=8$, $p=4$, $\alpha=0.1$. So prior $\blacksquare = 8/4 = 2$ per period.

After 5 zero periods, a demand of $y=10$ arrives ($q=5$ since last demand).

$$z_{\text{new}} = 8 + 0.1 \cdot (10 - 8) = 8.2$$

$$p_{\text{new}} = 4 + 0.1 \cdot (5 - 4) = 4.1$$

$$\blacksquare_{\text{new}} = 8.2/4.1 \approx 2.00 \text{ (almost flat here — small } \alpha \text{).}$$

$$\text{SBA } \blacksquare = (1 - 0.05) \cdot 2.00 = 0.95 \cdot 2.00 = 1.90 \text{ (slightly lower than Croston).}$$

4B.2 TSB worked numerical

WORKED NUMERICAL — TSB probability decay

Let $z=10$, $\pi=0.4$, $\alpha_{\text{prob}}=0.2$, $\alpha_{\text{size}}=0.2$. Prior $\blacksquare=4$.

Zero week: $\pi \leftarrow 0.4 + 0.2 \cdot (0 - 0.4) = 0.32$; z unchanged; $\blacksquare=3.2$.

Another zero: $\pi \leftarrow 0.32 + 0.2 \cdot (0 - 0.32) = 0.256$; $\blacksquare=2.56$.

Then $y=12$: $z \leftarrow 10 + 0.2 \cdot (12 - 10) = 10.4$; $\pi \leftarrow 0.256 + 0.2 \cdot (1 - 0.256) = 0.4048$; $\blacksquare \approx 4.21$.

Lesson: TSB turns down forecasts during long zeros faster than Croston's static z/p .

4B.3 WMAPE vs MASE — when each talks

Metric	Strength	Weakness
WMAPE	Intuitive % of demand volume missed	Can look great if few high-y weeks dominate
MASE	Comparable across SKUs; <1 beats naive	Needs enough history for naive scale

4B.4 Rolling origin evaluation (idea)

Step 2 style evaluation walks forward week by week: train on past, predict next week, score, advance. This matches how a pharmacy would actually use a weekly forecast. A single fixed split is simpler for LightGBM tabular training in Step 3, with metrics on the held-out tail — both appear in the project; know which artefact you are reading.

4B.5 Pinball loss intuition

For quantile τ : $\text{loss} = \tau \cdot (y - \blacksquare)$ if $y \geq \blacksquare$; else $(\tau - 1) \cdot (y - \blacksquare)$

$\tau=0.5 \rightarrow$ symmetric median; $\tau=0.95 \rightarrow$ heavy penalty if $\blacksquare$ too low

That is why q95 sits above q50: under-predicting the upper tail is costly in training.

4B.6 Ablation reading protocol

Compare `lgbm_env` vs `lgbm_noenv` WMAPE/MASE on the same test weeks, then vs SBA/TSB/MA. If `env` wins overall but loses on intermittent slice, say so — honesty > vanity. Quote `step3_metrics_summary.csv` / `step3_metrics_by_slice.csv` numbers from your run.

Step 3 feature $\rightarrow$ quantile heads

```
Feature vector (week t, SKU i):
[ lag1, lag2, ..., rolling means,
  rain_7, rain_14, pm25_*, alert_*, calendar_* ]
|
+--> LGBM q50 --> yhat / planning
+--> LGBM q90 --> sigma ingredient
+--> LGBM q95 --> sigma ingredient + stress
```

4B.7 Teacher commentary — common student mistakes

1. Calling LightGBM “neural net” — it is gradient-boosted trees.
2. Saying MAPE was used — it was deliberately excluded.
3. Reporting only overall WMAPE without pattern slices.
4. Claiming rain features are production-validated (violates D5).

MODULE 5 — INVENTORY SCIENCE (SS, ROP, FEFO, SIM)

5.1 Service level → Z

Definition — Cycle service level & Z

Cycle service level is the target probability of not stocking out during a replenishment cycle. For a normal demand approximation, Z is the standard normal quantile of that probability. At 95%, $Z \approx 1.645$.

$Z \approx 1.645$ for service level 95%

Configured in Step 4 defaults (--service-level 0.95). Higher service level → higher Z → more SS.

5.2 σ from quantile spreads

$$\sigma_{q95} = (q95 - q50) / Z_{0.95}$$

$$\sigma_{q90} = (q90 - q50) / Z_{0.90}$$

$$\sigma_{\text{weekly}} \approx \text{blend}(\sigma_{q95}, \sigma_{q90}) \text{ with floor } \approx 0.15 \times \mu$$

Implemented in src/safety_stock.py. Blending two spreads is more stable than one noisy tail.

5.3 Core inventory formulae

$$SS = Z \times \sigma_{LT} \times \sqrt{(L + R)}$$

$$ROP = \mu \times (L + R) + SS$$

Order-up-to $\approx$ ROP + review replenishment logic (see params CSV)

L = lead time (weeks), R = review period (weeks). Defaults L=R=1. μ = mean weekly demand (avg q50).

Symbol	Meaning	Typical DEV value
Z	Normal z for cycle service level	1.645 @ 95%
σ_{LT}	Demand uncertainty scale used with $\sqrt{(L+R)}$	From q90/q95 vs q50
L	Supplier lead time	1 week
R	Review period	1 week
μ	Mean demand per week	Avg q50
SS	Safety stock buffer	$Z \cdot \sigma \cdot \sqrt{(L+R)}$
ROP	Reorder when inventory position $\leq$ ROP	$\mu(L+R)+SS$

5.4 Worked numericals (practise all three)

WORKED NUMERICAL 1 — Basic SS and ROP

Given: $\mu = 10$ units/week, $\sigma_{LT} = 4$, $L = 1$, $R = 1$, service 95% ($Z = 1.645$).

Cover weeks = $L+R = 2$.

$SS = 1.645 \times 4 \times \sqrt{2} \approx 1.645 \times 4 \times 1.414 \approx 9.30$ units.

$ROP = 10 \times 2 + 9.30 = 29.30 \rightarrow$ order when on-hand + on-order $\leq \sim 29$.

If on-hand = 12 and nothing on order $\rightarrow$ replenish toward order-up-to.

WORKED NUMERICAL 2 — σ from quantiles then SS

Given weekly quantiles: q50 = 12, q90 = 22, q95 = 28. $Z_{0.95} = 1.645$, $Z_{0.90} \approx 1.282$.

$\sigma_{q95} = (28-12)/1.645 \approx 9.73$; $\sigma_{q90} = (22-12)/1.282 \approx 7.80$.

Blend (simple average for hand calc): $\sigma \approx (9.73+7.80)/2 \approx 8.77$. Floor $0.15 \times 12 = 1.8$ (not binding).

$L=R=1$, $Z=1.645 \rightarrow SS = 1.645 \times 8.77 \times \sqrt{2} \approx 1.645 \times 8.77 \times 1.414 \approx 20.4$.

$ROP = 12 \times 2 + 20.4 = 44.4$ units.

WORKED NUMERICAL 3 — Compare two service levels

Same $\mu=6$, $\sigma=3$, $L+R=2$. Compare 90% ($Z \approx 1.282$) vs 95% ($Z \approx 1.645$).

$SS_{90} = 1.282 \times 3 \times 1.414 \approx 5.44$; $ROP_{90} = 12 + 5.44 = 17.44$.

$SS_{95} = 1.645 \times 3 \times 1.414 \approx 6.98$; $ROP_{95} = 12 + 6.98 = 18.98$.

Lesson: higher service level buys fewer stockouts at the cost of more average inventory.

In sim KPIs, watch fill rate up vs waste/holding pressure.

WORKED NUMERICAL 4 — Inventory position trigger

$ROP = 40$. On-hand = 25, on-order = 10 $\rightarrow$ inventory position = $35 \leq 40 \rightarrow$ place PO.

If on-order were 20 $\rightarrow$ position = $45 > 40 \rightarrow$ do not double-order.

FEFO does not change ROP maths; it changes which lots are depleted and expiry waste.

5.5 FEFO vs FIFO — deep comparison

Basis	FIFO	FEFO
Pick order	Oldest receipt first	Earliest expiry first
Expiry waste	Can leave near-expiry untouched	Prioritises dying lots
Pharmacy fit	Generic warehouses / non-dated	Medicines with dated lots
Secondary key	Often receipt only	Expiry then receipt
In project	fifo_static baseline	fefo_* policies in Step 4
Risk if ignored	Hidden expiry write-offs	Slightly more complex picks

In `src/inventory_fefo.py`, `allocate_demand()` sorts lots by the policy, depletes `qty_on_hand`, and returns sold, unmet, revenue, COGS, markdown discount. FEFO reduces expiry write-offs when multiple lots coexist — common after overlapping purchases.

5.6 OTC markdown tiers

Days to expiry	Markdown (OTC eligible)	Notes
> 90	0%	Full price
61–90	10%	Early nudge
31–60	25%	Clearance begins
15–30	40%	Aggressive
< 15	50%	Last chance
Rx / H1 / habit-forming	0% auto	No automatic clinical discounting

Markdown applies only to eligible OTC classes (e.g. respiratory, pain analgesics, GI, vitamins, derma — as coded). Policy `fefo_ss_markdown` combines FEFO + SS reorder + these tiers and may uplift clearance demand in the simulator.

5.7 Simulation loop

FEFO simulation loop

```
Weekly sim loop (cloned state – live DB untouched):
1) receive any POs due this week
2) write off expired lots (waste KPI)
3) realise demand; allocate FEFO or FIFO
4) record sold / unmet / revenue / markdown
5) if inventory_position <= ROP: place PO arriving in L weeks
6) advance week; repeat over test horizon
```

5.8 Policy comparison — how to read results

Policy	What it adds	Typical DEV lesson
fifo_static	FIFO + static / no dynamic SS reorder	Baseline fill / waste
fefo_static	FEFO picks	Small waste improvement
fefo_ss	FEFO + probabilistic SS/ROP	Fill rate jumps
fefo_ss_markdown	OTC near-expiry discounts	Best waste+fill combo (DEV)

Documented DEV ballpark from step4_summary style runs: fifo fill rate often ~mid-80%; fefo_ss / fefo_ss_markdown can reach ~mid-90% fill. Quote your own step4_summary.json in the report — do not invent new digits.

Definition — Fill rate

Demand satisfied ÷ total demand (including unmet). Higher is better. Pair with waste/expiry KPIs — fill rate alone can hide overstock.

5.9 Isolation fingerprint

CAUTION / EXAMINER TRAP — Simulations fingerprint $\text{SUM}(\text{qty_on_hand})$ before/after. If it changes, the twin isolation invariant is broken — automatic acceptance failure.

5.10 Recommendations explanation text structure

Each REORDER recommendation typically carries: current_stock, forecast_demand, safety_stock, reorder_point, qty_suggested, and explanation_text that narrates the $\text{SS} = Z \cdot \sigma \cdot \sqrt{L+R}$ assumption set (service level, L, R). In demo, open Recommendations and read the explanation aloud — that is free viva credit.

MODULE 5B — INVENTORY DEEP DIVE

5B.1 Why $\sqrt{L+R}$ appears

If weekly demand deviations are roughly independent with std σ_{week} , variance over $H = L+R$ weeks adds: $\text{Var} = H \cdot \sigma_{\text{week}}^2$, so $\text{std} = \sigma_{\text{week}} \cdot \sqrt{H}$. Safety stock $Z \cdot \sigma \cdot \sqrt{L+R}$ is the classic normal approximation for the review/lead-time cover window. If your σ_{LT} is already defined over the cover horizon, do not multiply $\sqrt{H}$ twice — read `src/safety_stock.py` naming carefully in code review.

5B.2 Order-up-to narrative

Definition — Order-up-to level

A target inventory position after ordering. When position falls below ROP/trigger, order quantity $\approx \max(0, \text{order_up_to} - \text{position})$. `step4_safety_stock_params.csv` materialises per-SKU parameters including `order_up_to`.

5B.3 FEFO allocation micro-example

WORKED NUMERICAL — FEFO pick across two lots

Lots: A qty=5 expiry=2026-10-01; B qty=8 expiry=2026-11-15. Demand=7.

FEFO: take 5 from A, then 2 from B. A emptied (good — nearer expiry gone).

FIFO if A was received later than B could have left A sitting → higher expiry risk.

Waste KPI improves when near-expiry lots are systematically preferred.

5B.4 Markdown + clearance uplift (policy 4)

`fefo_ss_markdown` applies tier discounts on eligible OTC lots and may increase realised demand in the simulator for discounted offers (clearance uplift). That is a modelling choice to show how commercial levers interact with FEFO — not a claim that patients always buy more when vitamins are 25% off. Keep the tone engineering-honest.

5B.5 Reading `step4_policy_comparison.csv`

Column family	What you say in viva
Fill rate	% demand met — higher better
Unmet / stockout	Residual pain after policy
Expiry waste	Units/value written off
Orders placed	Replenishment activity under SS
Revenue / markdown \$	Commercial side-effect of tiers

CAUTION / EXAMINER TRAP — Do not memorise a single fill-rate percentage from an old chat. Open your `step4_summary.json` and quote that run.

5B.6 Recommendations text — template anatomy

Recommendation narrative anatomy

```
explanation_text roughly narrates:
"Demand  $\mu=.$ ; sigma $=.$ ; service 95% ( $Z=1.645$ );
L=1,R=1; SS= $Z \cdot \sigma \cdot \sqrt{L+R}=.$ ; ROP= $\mu \cdot (L+R) + \text{SS}=.$ ;
on-hand $=.$ ; suggest order qty $=.$ ."
Desktop Recommendations page: filter action_type=REORDER and read aloud.
```

MODULE 6 — REGULATED SUBSTITUTE ENGINE

Definition — Pharmacist-reviewed substitute

An inventory alternative ranked by composition/class similarity (TF-IDF + cosine), then filtered by Indian regulatory hard gates. Output is for pharmacist review — not automatic clinical prescribing (D4).

6.1 Explicit catalog substitutes vs TF-IDF

Path	How it works	When used
Explicit	Catalog columns substitute0–substitute4 if in stock	Prefer when curated links exist
TF-IDF fallback	Vectorise composition/chemical/therapeutic text; cosine rank	When explicit list empty / insufficient

6.2 TF-IDF + cosine similarity

$$\text{cosine}(a,b) = (a \cdot b) / (||a|| ||b||)$$

a,b = TF-IDF vectors of composition / class text

Higher cosine $\Rightarrow$ more similar salt / therapeutic tokens. Default min_cosine $\approx$ 0.55 in Step 5.

TF (term frequency) upweights tokens that appear often in a medicine's composition text; IDF downweights tokens that appear in almost every medicine (poor discriminators). Vectors are L2-normalised so cosine equals a simple dot product in the engine.

6.3 Full gate logic

Substitute pipeline

```
Query SKU
-> gather explicit substitutes + TF-IDF neighbours
-> keep in-stock candidates only
-> GATE pipeline (any fail => block candidate or whole query):
  Schedule H1? -> hard block + register/Rx notice
  AWaRe Watch/Reserve antibiotic path? -> block OTC auto-sub
  CDSCO banned FDC / discontinued? -> hard block
  Habit-forming flag? -> block
-> return top-N allowed OR single H1 block message
-> pharmacist reviews (D4) – system does not dispense
```

Gate	Action	Fresher meaning
Schedule H1	Hard block + notice	Strict Rx; 3-year dispensing record duty
AWaRe Access	Allowed (stewardship-friendly)	Prefer if antibiotic path
AWaRe Watch/Reserve	Block auto-sub	Higher AMR risk
CDSCO banned FDC	Hard block	Irrational / withdrawn combinations
Discontinued	Hard block	Is_discontinued flag
Habit forming	Hard block	No casual substitution

Lists are curated project subsets for demo coverage — say that honestly in viva (see Module 11 limitations). WHO AWaRe and CDSCO are real frameworks; the embedded pattern tables are not the entire national database.

6.4 Pharmacist-reviewed only (D4)

Four points examiners look for — substitute ethics

1. UI label is Substitutes (review) — not “auto switch”.
2. No dose/diagnosis logic — inventory alternatives only.
3. H1 block must show educational notice, not a silent empty list.
4. Final dispensing decision remains with the registered pharmacist.

6.5 Worked example — blocked vs allowed

WORKED EXAMPLE — Gate outcomes

Case A — Query is Schedule H1 antibiotic: engine returns HARD BLOCK with RMP Rx / register notice. Even high-cosine neighbours are not offered for auto-sub.

Case B — Query is OTC vitamin, candidate same salts, cosine 0.78, in stock, not habit-forming, not banned FDC: ALLOWED $\rightarrow$ listed in top-N for pharmacist review.

Case C — Candidate is WHO AWaRe Watch while query path requires stewardship gate: BLOCKED despite high cosine.

Case D — Explicit substitute listed but out of stock: skipped; TF-IDF may fill.

6.6 Audit metrics meaning

Metric (Step 5)	Meaning
n queries	Stockout substitute lookups run in batch audit
% with ≥ 1 allowed	How often the engine finds a legal in-stock alt
H1 fully blocked	Queries where H1 gate stopped suggestions
Flag counts H1/Watch/Reserve	Assortment regulatory annotation volume

Example DEV summary style (quote your step5_summary.json): on the order of 200 queries, majority with ≥ 1 allowed substitute, dozens of H1 fully blocked — exact numbers come from your run artefacts, not from memory.

MODULE 7 — DESKTOP APPLICATION & OPERATIONS

7.1 Launch and navigation

Entry: `py -3 scripts/run_desktop.py` → `app/main.py`. Tkinter + ttk pages. Nav also exposes Set MySQL password, Seed database, Load analytics, Refresh Overview.

7.2 Every page — what to demonstrate

Page	Student must demonstrate	Pass signal
Overview / Twin	Catalog ~254k vs stocked ~3k; synthetic banner; twin sync	<code>data_mode=dev_synthetic</code> visible
Inventory	Search working meds; FEFO lots; add clone; add/remove lot	Reference not editable
Forecasts	Pick SKU → weekly $\hat{y}$ /q50/q95 from MySQL	Rows present after <code>load_analytics</code>
Recommendations	Filter REORDER; read SS formula explanation	<code>explanation_text</code> non-empty
Alerts	Near-expiry ≤90d, low stock	Open alerts list
Simulations	Import cached Step 4 / what-if; stock unchanged	Fingerprint OK
Substitutes	Allowed list OR H1 block message	No auto-dispense claim
Settings	Password, seed, load analytics	Probe succeeds

7.3 Add / remove inventory fields

Action	Key fields / behaviour
Add medicine (clone)	Search reference → copy name/generic/schedule/form → new working row
Add lot	<code>batch_no</code> , <code>mfg_date</code> , <code>expiry_date</code> , <code>qty_on_hand</code> , <code>qty_unit</code>
Remove lot / med	Working sources only; audited; reference blocked
Guard	<code>InventoryGuardError</code> on reference mutation attempts

7.4 Demo script / acceptance checks

`py -3 scripts/run_demo_acceptance.py` runs the automated checklist (documented as 21 checks in `docs/acceptance_checklist.md`). Manual walkthrough still required for UI clarity. Targets include catalog ≥ ~250k, stocked toward 3000, synthetic banner, sim isolation.

7.5 Operator commands cheat sheet

Operator cheat sheet

```
cd e:\MajorProject_4th_Yr\Agent
py -3 scripts/run_step1.py
py -3 scripts/run_step2.py
py -3 scripts/run_step3.py
py -3 scripts/run_step4.py
py -3 scripts/run_step5.py
py -3 scripts/seed_mysql.py --apply-schema
py -3 scripts/apply_migration_002.py
py -3 scripts/load_analytics.py
py -3 scripts/snapshot_originals.py
py -3 scripts/run_demo_acceptance.py
py -3 scripts/run_desktop.py
```

7.6 Common failure modes

Symptom	Likely cause	Fix
DB connection fails	Wrong MySQL password / server down	Settings password; check service
Empty forecasts page	Analytics not loaded	<code>load_analytics.py</code> after Steps 2–3
Tiny catalog count	Seed without schema/files	<code>run_step1</code> then <code>seed --apply-schema</code>
Cannot edit medicine	It is a reference row	Clone to working stock first
Substitutes empty	Gates blocked or no stock	Check H1 notice; verify lots

Symptom	Likely cause	Fix
Sim changes stock	Isolation bug / wrong service	Fail acceptance; do not demo
ImportError / missing parquet	Pipeline not run	run_step1..5 in order
CAUTION / EXAMINER TRAP — Wrong MySQL password is the #1 lab-day failure. Missing load_analytics is #2 (seed alone may not refresh forecast/recommendation tables the way you expect).		

MODULE 6B–7B — SUBSTITUTES & OPS DEEP DIVE

6B.1 composition_text() role

Before TF-IDF, catalog helps concatenate chemical composition / therapeutic class tokens into a normalised string. Garbage in (empty composition) $\Rightarrow$ weak cosine neighbours. That is why explicit substitute0–4 columns remain valuable when curated.

6B.2 min_cosine threshold trade-off

min_cosine	Effect
Too low (e.g. 0.2)	Noisy unrelated suggestions — dangerous UX
Default ~0.55	Project balance for demo precision
Too high (e.g. 0.9)	Many empty results even when reasonable alts exist

6B.3 Schedule H1 teaching points (India retail)

Schedule H1 drugs require prescription from a Registered Medical Practitioner and maintenance of supply records for a statutory period (commonly taught as three years). The engine's hard block + notice is an educational/compliance UX pattern for the twin — not a full e-register product.

7B.1 Demo day runbook (minute-by-minute idea)

Minute	Action
0–2	Overview: catalog vs stocked + synthetic banner
2–5	Inventory: show FEFO lots; clone add if time
5–8	Forecasts + Recommendations explanation_text
8–11	Simulations: import Step 4; prove stock unchanged
11–14	Substitutes: one allowed + one H1 block
14–15	Limitations sentence (D3/D4/D5/D7)

7B.2 Seed vs load_analytics — decide which

Full seed_mysql rebuilds reference+working+transactions (slow, heavy). load_analytics refreshes forecast/recommendation flavoured tables from processed artefacts when inventory already exists. Lab tip: after changing only Step 3/4 outputs, prefer load_analytics.

WORKED OPS SCENARIO — empty Forecasts page on demo morning

Symptom: Forecasts page blank; Overview shows 254k/3k OK.

Diagnosis: seed ran, but analytics not loaded / Step 3 artefacts missing.

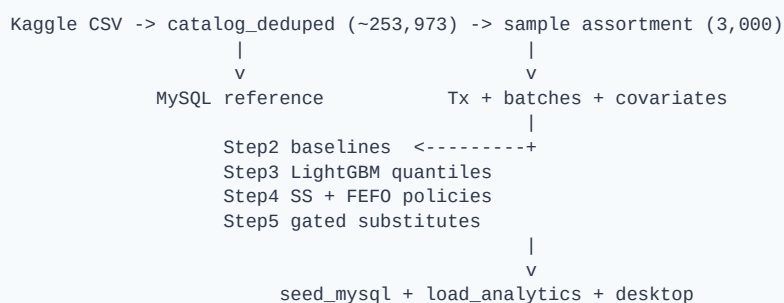
Fix: ensure step3_forecasts_weekly exists $\rightarrow$ py -3 scripts/load_analytics.py $\rightarrow$ Refresh.

Prevention: run_demo_acceptance.py before the panel arrives.

MODULE 8 — PIPELINE STEPS 1–5 END-TO-END

8.0 Master pipeline diagram

Pipeline redraw for viva



8.1 Step 1 — Synthetic world build

Step 1 I/O

```
run_step1.py -> src/generate_transactions.py
IN: India A-Z CSV
OUT: catalog_deduped.parquet (~253,973)
    assortment_3k.csv (~3,000)
    covariates_navi_mumbai.csv
    transactions_synthetic.*
    batches_fefo.*
    step1_summary.json
```

Also produces counts for days, transactions, units sold, stockout events, batches, revenue in step1_summary.json — cite that file rather than inventing figures.

8.2 Step 2 — Intermittent baselines

Step 2 I/O

```
run_step2.py -> src/run_forecast_eval.py
Methods: croston, sba, tsb, ma
Metrics: WMAPE, MASE (+ pattern slices)
OUT: step2_metrics_by_sku.csv
    step2_metrics_summary.csv
    step2_forecasts_weekly.*
    step2_summary.json
```

8.3 Step 3 — LightGBM quantiles + env ablation

Step 3 I/O

```
run_step3.py -> src/run_env_forecast.py
Models: lgbm_env & lgbm_noenv @ q50/q90/q95
OUT: step3_forecasts_weekly.* (feeds Step 4)
    step3_metrics_summary.csv
    step3_feature_importance.csv
    models_step3/*.txt
    step3_summary.json
```

8.4 Step 4 — Safety stock + FEFO policy simulation

Step 4 I/O

```
run_step4.py -> src/run_inventory_sim.py
Policies: fifo_static | fefo_static | fefo_ss | fefo_ss_markdown
OUT: step4_safety_stock_params.csv
    step4_policy_comparison.csv
    step4_weekly_kpis.csv
    step4_sim_detail.parquet
    step4_summary.json
```

8.5 Step 5 — Substitutes audit

Step 5 I/O

```
run_step5.py -> src/run_substitutes.py
OUT: step5_assortment_regulatory.csv
    step5_stock_snapshot.csv
    step5_substitute_recommendations.csv
    step5_audit_counts.csv
    step5_summary.json
```

8.6 Seed, migrate, analytics, snapshot

Script	Role
seed_mysql.py --apply-schema	Create DB/tables; load reference+working+Tx
apply_migration_002.py	Editable inventory columns + audit
load_analytics.py	Refresh forecasts/recommendations without full reseed
snapshot_originals.py	Freeze processed artefacts to data/originals/

8B.1 Step 1 acceptance numbers to verify

Open `data/processed/step1_summary.json` after a run. Expect `catalog_deduped` $\approx$ 253,973, `assortment` $\approx$ 3,000, multi-hundred-thousand synthetic transactions scale, thousands of batches. If `assortment` is 50, you passed the wrong CLI flag — do not demo.

8B.2 Step 2 α/β knobs

CLI exposes `--alpha/--beta` style smoothing parameters for intermittent methods. Smaller α = smoother, slower reaction; larger α = jumpy. Default 0.1 is a teaching compromise. Do not retune live in viva unless asked.

8B.3 Step 3 model files

`models_step3/lgbm_env_q50.txt` (and `q90/q95`, plus `noenv` twins) are LightGBM text model dumps. They prove training happened and support reload experiments. The desktop does not need these files if forecasts are already in MySQL.

8B.4 Step 4 CLI defaults worth memorising

```
--lead-time-weeks 1 --review-period-weeks 1 --service-level 0.95
```

These match the SS numericals in Module 5. Changing them changes ROP — say so if you change them.

8B.5 Step 5 CLI defaults

Typical: `--n-queries 200 --top-n 5 --min-cosine 0.55`. Audit CSVs let you show gated vs allowed counts without clicking the UI fifty times.

MODULE 9B — EXTENDED GLOSSARY & COMPARISON SHEETS

Term	One-line definition
Pinball loss	Asymmetric quantile training loss
Ablation	Remove a feature family to measure lift
Rolling origin	Walk-forward one-step evaluation
Stratified sample	Preserve cohort proportions when choosing 3k
Immutability	Reference catalog must not be UI-edited
Fingerprint	Checksum-like SUM(qty) isolation check
Order-up-to	Target position after replenishment
Clearance uplift	Simulated demand boost under markdown
Stewardship	Antibiotic responsibility (AWaRe)
FDC	Fixed-dose combination
RMP	Registered Medical Practitioner
Grain W	Weekly forecast grain in MySQL
ttk	Themed Tkinter widgets used in UI
Parquet	Columnar file format for pipeline outputs
Acceptance pack	Automated + manual demo checklist
Stale twin	Analytics/inventory sync lag signal
Working copy	Editable stocked medicine row
Explicit substitute	Curated substitute0–4 link
Hard gate	Non-negotiable regulatory block
Soft rank	Cosine ordering before gates

9B.1 Method cheatsheet table

Method	Updates on zeros?	Core formula idea	File
Croston	No (z,p frozen)	z/p	forecast_intermittent.py
SBA	No	$(1-\alpha/2)z/p$	same
TSB	Yes (π decays)	$\pi \cdot z$	same
MA	Yes (window mean)	mean(window)	same
LGBM $q\tau$	N/A (tabular)	pinball@ τ	forecast_env_lgbm.py

MODULE 10B — MORE VIVA DRILLS (SHORT HINTS)

Q1 (extra). Explain dual store in 20 seconds.

Hint: Parquet science files + MySQL twin state; load_analytics bridges.

Q2 (extra). What breaks if unmet_qty ignored?

Hint: Stockouts look like zero demand; SS/FC collapse.

Q3 (extra). Draw FEFO vs FIFO on two lots.

Hint: Sort expiry vs receipt; allocate demand.

Q4 (extra). Why fingerprint sims?

Hint: Prove isolation invariant; acceptance gate.

Q5 (extra). Difference D4 vs clinical CDSS?

Hint: Inventory alt review vs diagnosing/prescribing.

Q6 (extra). What is pinball@0.95 doing?

Hint: Punish under-prediction of upper demand tail.

Q7 (extra). Name migration 002 artefacts.

Hint: form_type, qty_unit, pharmacy source, inventory_audit.

Q8 (extra). When use seed vs load_analytics?

Hint: Seed rebuilds world; load refreshes FC/recs.

Q9 (extra). What does step4_safety_stock_params hold?

Hint: Per-SKU $\mu, \sigma, SS, ROP, order_up_to, z, \dots$

Q10 (extra). How does Overview know synthetic?

Hint: app_meta data_mode + twin service banner.

Q11 (extra). Why stratified 3k?

Hint: Keep intermittent cohorts; avoid only fast movers.

Q12 (extra). SBA correction factor at $\alpha=0.2$?

Hint: $1-\alpha/2=0.9$.

Q13 (extra). Cosine of identical L2 vectors?

Hint: 1.0

Q14 (extra). Cosine of orthogonal TF-IDF?

Hint: 0.0

Q15 (extra). What is WRITEOFF_EXPIRY?

Hint: stock_movements type for expired lot removal.

Q16 (extra). Name services/simulations promise.

Hint: No live batch mutation.

Q17 (extra). Why q90 and q95 both?

Hint: Blend more stable σ than one noisy quantile.

Q18 (extra). What is habit-forming gate?

Hint: Block auto-sub candidates flagged habit forming.

Q19 (extra). Demo order of pages?

Hint: Overview→Inventory→FC/Recs→Sim→Subs→limitations.

Q20 (extra). Where are decisions locked written?

Hint: docs/decisions_locked.md

Q21 (extra). External catalog in git?

Hint: No — large CSV lives under MajorProject Data path.

Q22 (extra). What is STALE_TWIN alert?

Hint: Twin sync/analytics freshness warning type.

Q23 (extra). Fill rate formula?

Hint: Satisfied / (satisfied+unmet).

Q24 (extra). Why not Streamlit-first?

Hint: D2 pharmacy desktop expectation.

Q25 (extra). Can markdown apply to H1?

Hint: No automatic OTC-style markdown on restricted Rx paths.

MODULE 11B — ETHICS SCENARIOS & REPORT LANGUAGE

11B.1 Language that is safe vs unsafe

Unsafe claim	Safe rewrite
"Trained on Bhagyashree sales"	"Trained on labeled dev_synthetic demand"
"AI prescribes alternatives"	"Suggests inventory alternatives for pharmacist review"
"Rain causes antibiotic sales"	"Rain lags are exploratory features (D5)"
"Detects anemia in patients"	"Anemia product gated (D7); not shipped"
"MAPE 12% proves accuracy"	"WMAPE/MASE on intermittent panel"

11B.2 Scenario drills

SCENARIO A — Examiner: "Show me patient diagnosis."

Reply: Out of scope. Twin manages inventory/demand decisions, not clinical diagnosis (D4/D7/D8).

Offer: Forecasts, SS recommendations, FEFO sim, gated substitutes instead.

SCENARIO B — Examiner: "Is FEFO always better?"

Reply: For dated medicine lots, FEFO targets expiry waste; FIFO may match if single lot.

Evidence: compare fefo_* vs fifo_static KPIs in step4_policy_comparison.

SCENARIO C — Examiner: "Prove you did not touch live stock in sim."

Reply: Isolation fingerprint on SUM(qty_on_hand); simulations write simulation_* tables only.

Demo: note on-hand before/after what-if on Simulations page.

11B.3 One-page oral summary (60 seconds)

PharmTwinAI is a MySQL-backed desktop digital twin for retail pharmacy operations in a Navi Mumbai context. We keep ~253,973 reference medicines immutable and ~3,000 working SKUs editable. Offline Steps 1–5 build synthetic intermittent demand, score Croston/SBA/TSB and LightGBM quantiles with WMAPE/MASE, compute SS/ROP, compare FEFO policies, and audit gated substitutes. The UI explains reorders and never auto-prescribes. Data is dev_synthetic until sponsor POS arrives; anemia screening stays gated.

MODULE 9 — GLOSSARY, ACRONYMS, FORMULA SHEET

9.1 Large glossary

Term	One-line definition
SKU	Stock-keeping unit — sellable pack identity
Lot / batch	Physical stock sharing mfg & expiry
Assortment	Active shelf subset (~3,000) of full catalog
Reference catalog	Immutable ~253,973 knowledge base rows
Working stock	Editable pharmacy/dev_synthetic/sponsor rows
Digital twin	Software mirror of pharmacy operational state
Intermittent demand	Many zero periods + occasional spikes
ADI	Average demand interval between non-zero sales
CV ²	Squared coefficient of variation of demand sizes
Croston	Smooth size & interval separately
SBA	Syntetos–Boylan bias-corrected Croston
TSB	Teunter–Syntetos–Babai probability×size forecast
WMAPE	$\frac{\sum y_t - \hat{y}_t }{\sum y_t }$ — scale-free intermittent-safe
MASE	Error scaled by naive lag-1 absolute changes
MAPE	Forbidden headline metric here (zero blow-up)

Term	One-line definition
Pinball loss	Quantile regression training loss
q50 / q90 / q95	Predictive demand quantiles
Service level	Target non-stockout probability in a cycle
Z	Normal quantile of service level (~1.645 @ 95%)
Safety stock (SS)	$Z \times \sigma \times \sqrt{(L+R)}$ buffer
ROP	$\mu(L+R) + SS$ reorder trigger
Lead time L	Wait until PO arrives
Review period R	How often stock position is reviewed
Inventory position	On-hand + on-order – backorders
Fill rate	Satisfied demand ÷ total demand
Stockout	Demand with no sellable quantity
Unmet qty	Recorded lost demand during stockout
FEFO	First-Expiry-First-Out picking
FIFO	First-In-First-Out picking
Markdown tier	OTC near-expiry discount schedule
TF-IDF	Term weighting for text importance
Cosine similarity	Angle-based vector similarity in [0,1] after L2
Schedule H1	Strict Indian Rx schedule with register duties
WHO AWaRe	Access / Watch / Reserve antibiotic groups
CDSCO	Indian drug regulator; banned FDC lists
dev_synthetic	Labeled DEV data mode — not live POS
Twin isolation	Sims must not mutate live batch qty
Clone	Insert working medicine copied from reference

9.2 Acronyms quick list

WMAPE, MASE, MAPE, ADI, CV, SS, ROP, FEFO, FIFO, OTC, TF-IDF, LGBM/LightGBM, AWaRe, CDSCO, FDC, RMP, PO, KPI, NDA, SKU, UI, DB, CRUD, AMR.

9.3 Formula sheet summary (copy to one A4)

$$WMAPE = \sum |y - \hat{y}| / \sum |y|$$

$$MASE = \text{mean}(|e|) / \text{mean}(|y_t - y_{t-1}|)$$

$$SBA \alpha = (1 - \alpha/2)(z/p) \quad TSB \alpha = \pi \cdot z$$

$$\sigma_{\alpha} \approx \text{blend}((q95 - q50)/Z0.95, (q90 - q50)/Z0.90)$$

$$SS = Z \cdot \sigma \cdot \sqrt{(L+R)} \quad ROP = \mu(L+R) + SS \quad Z \approx 1.645 @ 95\%$$

$$\text{cosine}(a, b) = (a \cdot b) / (||a|| \cdot ||b||) \text{ on TF-IDF vectors}$$

NEVER MAPE for intermittent demand. Demand y includes unmet_qty in this project.

MODULE 10 — VIVA / PROBABLE QUESTIONS (40+)

Each item: question, then a **short hint** (not a full essay). Practise answering aloud in under a minute. Expand only if the examiner probes.

10.1 Question bank with hints

Q1. What is PharmTwinAI in one sentence?

Hint: Desktop pharmacy digital twin: stock+forecast+SS/FEFO sim+gated substitutes; MySQL+tkinter.

Q2. Why call it a twin not a dashboard?

Hint: Holds decision state + what-if isolation, not only charts.

Q3. What is D2?

Hint: Desktop-first (tkinter), not Streamlit-first.

Q4. What is D3?

Hint: Synthetic labeled data until sponsor POS; honesty banner.

Q5. What is D4?

Hint: Pharmacist-reviewed substitutes only — no auto-prescribe.

Q6. What is D5?

Hint: Env features research/DEV until proven on real sales.

Q7. What is D7?

Hint: Anemia screening gated — not in core product yet.

Q8. Catalog size?

Hint: ~253,973 deduped reference SKUs.

Q9. Working assortment size?

Hint: ~3,000 SKUs.

Q10. Can Inventory UI edit reference rows?

Hint: No — clone to working; InventoryGuardError.

Q11. What does source_system distinguish?

Hint: reference vs dev_synthetic/pharmacy/sponsor.

Q12. Why intermittent methods?

Hint: Many zero weeks; separate size vs interval/probability.

Q13. Croston in one line?

Hint: Smooth z and p ; $\hat{z} = z/p$; update on positive demand.

Q14. What does SBA fix?

Hint: Croston positive bias via $(1-\alpha/2)$ factor.

Q15. TSB vs Croston?

Hint: TSB updates π every period including zeros; $\hat{\pi} = \pi Z$.

Q16. Write WMAPE.

Hint: $\sum |y - \hat{y}| / \sum |y|$.

Q17. Write MASE idea.

Hint: Mean abs error / mean abs naive lag-1 change.

Q18. Why not MAPE?

Hint: Zeros undefined/explode; purple numerical blow-up.

Q19. What is unmet_qty for?

Hint: Capture stockout demand so FC sees latent demand.

Q20. LightGBM quantiles used?

Hint: q50, q90, q95 with pinball/quantile objective.

Q21. How is σ estimated?

Hint: Blend $(q95 - q50)/Z0.95$ and $(q90 - q50)/Z0.90$; floor.

Q22. Write SS formula.

Hint: $Z \cdot \sigma \cdot \sqrt{(L+R)}$.

Q23. Write ROP formula.

Hint: $\mu(L+R)+SS$.

Q24. Z at 95%?

Hint: ≈ 1.645 .

Q25. Default L and R?

Hint: 1 week each in Step 4 defaults.

Q26. FEFO vs FIFO?

Hint: Earliest expiry vs earliest receipt; pharmacy prefers FEFO.

Q27. Name four sim policies.

Hint: `fifo_static`, `fefo_static`, `fefo_ss`, `fefo_ss_markdown`.

Q28. What KPI pair to read?

Hint: Fill rate and expiry waste — not fill alone.

Q29. Prove sim isolation.

Hint: Fingerprint `SUM(qty_on_hand)` unchanged.

Q30. OTC markdown idea?

Hint: Tiered discounts as expiry nears; Rx/H1 not auto-markdown.

Q31. Substitute ranking?

Hint: Explicit cols then TF-IDF cosine on composition text.

Q32. Write cosine formula.

Hint: $(a \cdot b) / (||a|| ||b||)$.

Q33. Schedule H1 gate?

Hint: Hard block + Rx/register notice.

Q34. AWaRe Watch/Reserve?

Hint: Block casual auto-sub on antibiotic stewardship path.

Q35. CDSCO gate?

Hint: Block banned FDC / irrational combinations in curated list.

Q36. Offline vs online?

Hint: Steps 1–5 train/export; UI reads MySQL via services.

Q37. Path of a click?

Hint: `app/main` → `services` → MySQL.

Q38. `load_analytics` purpose?

Hint: Refresh forecasts/recs without full reseed.

Q39. migration 002 adds?

Hint: `form_type`, `qty_unit`, pharmacy source, `inventory_audit...`

Q40. `snapshot_originals`?

Hint: Freeze processed files to `data/originals/`.

Q41. Env feature examples?

Hint: `rain>20mm` lags, PM2.5, health alerts + demand lags.

Q42. Feature importance caveat?

Hint: Importance $\neq$ causation (D5).

Q43. Who is the industry context?

Hint: Bhagyashree Medical, Juinagar, Navi Mumbai — context only; DEV data synthetic.

Q44. Is this a chatbot?

Hint: No — inventory/decision twin.

Q45. Acceptance endpoint?

Hint: `scripts/run_demo_acceptance.py` (~21 checks) + manual UI walk.

Q46. What if MySQL password wrong?

Hint: Connection fail — fix in Settings / `.env`.

Q47. Empty Forecasts page?

Hint: Run pipeline + `load_analytics`.

Q48. Order-up-to vs ROP?

Hint: ROP triggers; order-up-to is target position in params.

Q49. Why weekly grain?

Hint: Retail planning + intermittent stability vs noisy daily zeros.

Q50. Pinball loss one-liner?

Hint: Asymmetric loss targeting a quantile.

Q51. What NOT to claim?

Hint: Real POS used; auto-prescribe; proven weather causality; anemia product done.

Count: 51 probable questions with hints. Add your own from the latest step*_summary.json numbers before the final viva.

11.1 Technical limitations (say these unprompted)

1. DEV demand is **synthetic** — metrics are for method comparison, not published clinical/ops claims on Bhagyashree sales.
2. Regulatory lists (H1 patterns, AWARe subset, CDSCO banned FDCs) are **curated subsets** for the project, not exhaustive national databases.
3. Normal SS approximation and quantile-derived σ are practical engineering choices, not a full Bayesian inventory theory thesis.
4. Desktop+localhost MySQL fits D6 lab constraint; not a multi-store cloud SaaS yet.
5. TF-IDF substitutes use text similarity — they are not clinical equivalence certificates.

11.2 Ethics & professional boundaries

Definition — Human-in-the-loop dispensing

PharmTwinAI may suggest inventory alternatives and show regulatory notices, but a registered pharmacist remains responsible for legality and appropriateness of dispensing (D4).

CAUTION / EXAMINER TRAP — Never demo the system as diagnosing disease, recommending therapy, or replacing a pharmacist. Never present synthetic sales as sponsor POS.

11.3 Gated anemia & other screening (D7/D8)

Anemia-related research notes may exist in docs, but **product implementation is gated until core twin acceptance (D7)**. Other screening ideas are research-only until explicit per-condition approval (D8). In viva: core = catalog/stock twin + forecast + SS/FEFO + gated substitutes.

11.4 Future work (realistic)

Theme	Next step when unlocked
Sponsor POS	Replace synthetic Tx; keep labels honest during transition
Env validation	Re-test D5 features on real sales; drop if no lift
UI toolkit	Optional PySide6 migration keeping D2 desktop
Regulatory packs	Expand gated lists with pharmacist-reviewed sources
Multi-store	Only after local single-store acceptance + data agreements
Screening modules	Only after D7/D8 unlock criteria

11.5 Closing paragraph for reports

Closing line: PharmTwinAI is a desktop pharmacy digital twin that combines intermittent-demand forecasting (WMAPE/MASE; Croston/SBA/TSB; LightGBM quantiles), probabilistic safety stock and FEFO policy comparison in isolated simulation, and legally gated substitute suggestions for pharmacist review — with a clear separation between an immutable national medicine catalog (~253,973) and editable working stock (~3,000), and with explicitly labeled **dev_synthetic** data until sponsor POS arrives.

Four points examiners look for — final honesty checklist before you submit

1. Named methods present in report + viva.
2. Green formulae correct; MAPE not claimed.
3. Diagram redrawn once from memory.
4. Limitations + D3/D4/D5/D7 spoken aloud once.